ACPI Support for RISC-V RERI: Proposal of Work

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Agenda

- **Background**
- **Why Existing Extensions Are Insufficient?**
- **Does this work need to be done at RVI Foundation?**
- **Objectives**
- **Milestone**

Background: RAS Error handling philosophy

- Two methods for handling RAS errors
  - Firmware first Mode (FFM)
    - Kernel is prohibited from accessing the RAS error records directly. Firmware creates CPER records for Kernel to navigate and process
  - Kernel first Mode (KFM)
    - Kernel navigates the std error records directly
- How to use
  - In general, FFM can satisfy the requirements very well. It can hide implementation details from the OS, and it offers good scalability and practicality.
  - However, in large-scale data centers, KFM—especially CE collection in KFM mode—is required for widely used error prediction.

Background: Memory Predictive Failure Analysis Large Scale Clouds

- Key findings: CE samples play a crucial role in UE prediction
  - Most (58%~92%) DIMMs UEs have experienced preceding CEs in history[3]
  - Over 98% of DRAM faults map to a single bank on DRAM
  - 63% of nodes encounter DRAM errors from a single VM, while 90.9% experience errors from no more than three VMs.
- How to use
  - CE samples include error patterns, row/column characteristics, and ECC-related properties.
  - Prioritize collecting each CE error whenever possible, instead of relying on threshold-based reporting.
  - Use enough CE samples to train UE prediction models.
- Benefits
  - Avoids more than 10% of VM crashes [1]
  - Reduce memory error rate by 67% [2]

[1] Predicting DRAM-Caused Risky VMs in Large-Scale Clouds
[2] Revisiting Memory Errors in Large-Scale Production Data Centers: Analysis and Modeling of New Trends from the Field
[3] Predicting Uncorrectable Memory Errors from the Correctable Error History

Background: FFM Limitations in CE collection

- Correctable Errors (CE), occur more often than UE.
  - CE frequency is 10–1000× higher than UE frequency.[1]
  - CE storm: thonds of CE in s
- Field Replaceable Unit : CE system address need to translate to DIMM address, e.g. row, column, bank, to identify faulty DIMM.
- As a result: FFM introduce more context switch and leading cycle stealing(error report and error decode)

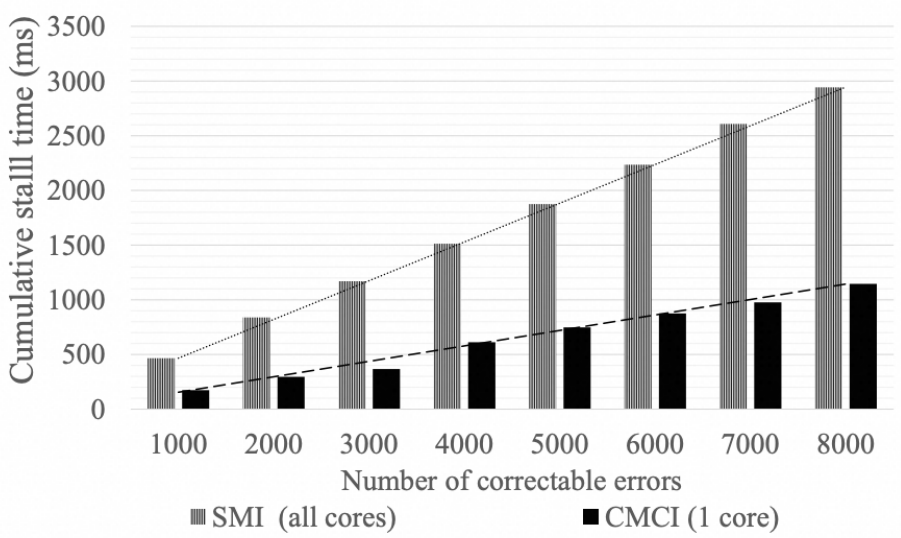


Figure 8: Comparison between SMI and CMCI stalls.

Optimizing Interrupt Handling Performance for Memory Failures in Large Scale Data Centers
SMI: FFM vs CMCI: KFM

From a performance perspective, KFM is a more suitable CE collection scheme for use in data centers.

[1] DRAM Errors in the Wild: A Large-Scale Field Study

Why Existing Extensions Are Insufficient?

Existing Extentions： Enough for FFM

- RERI(RISC-V Error Resource Interface)[1]: what information an error source reports
- SSE(Supervisor Software Events)[2]: how Firmware notify kernel
- Hardware Error Trap[3]: How Hardware Notify Software

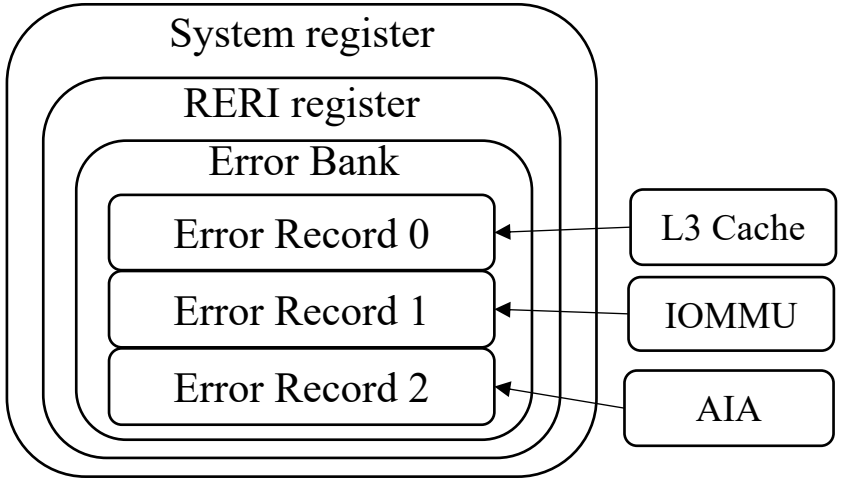
KFM need:

- Error register interface:
  - RERI ✅
- Error resource runtime discovery: how software discovers or binds to those resources
  - ❓
- Error decoding:
  - driver decoding, ACPI PRMT

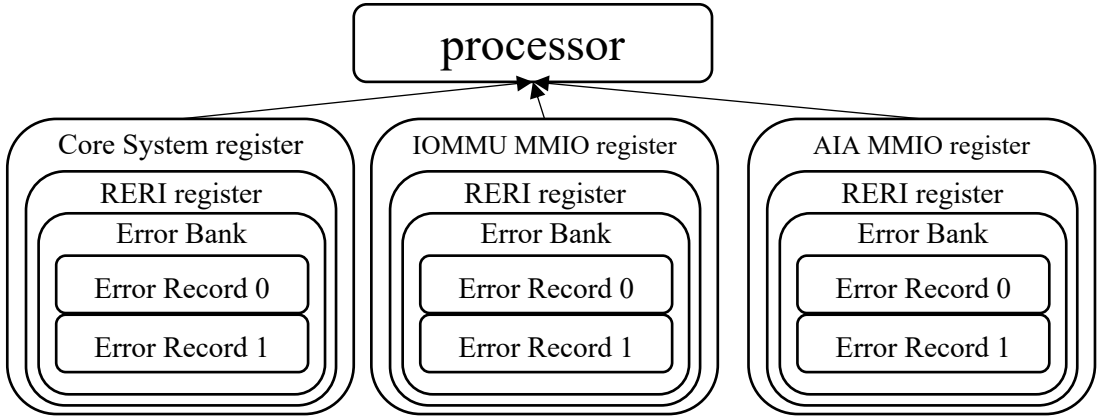
[1] https://github.com/riscv-non-isa/riscv-ras-eri
[2] https://github.com/riscv-non-isa/riscv-sbi-doc/blob/master/src/ext-sse.adoc
[3] https://github.com/riscv/riscv-isa-manual

Does this need to be done at RVI?

The hardware can be implemented with different register layouts. Two simple examples are shown below:



System register view



System/MMIO register view

Regardless of the hardware implementation, the information required by the software is largely the same and consistent: the mapping between the error bank/record and the error source, as well as which interrupt is used

- In FFM, this is not a problem, because firmware is typically delivered to users along with the hardware and almost not replaced, so vendors can use their own implementation approach.
- However, in KFM, a unified approach is needed to avoid fragmentation in kernel drivers. ACPI was introduced to address this type of issue, and RVI is the best platform to handle it.

Objectives

An architecture-specific ACPI table: the RISC-V Error Bank Table (RBRT)

- Error bank access model
  - System register or MMIO.
- Error resource mapping
  - Which error resource this error bank belongs to, cache?Interconnect mesh?AIA?IOMMU?
- Error resource interrupt
- Vendor-specific framwork
- Error decoding(Optional)
  - system address to DIMM address

Milestones

- Discuss in PRS TG
- Specification draft
- QEMU prototype
- Linux driver prototype
- Specification review and ratification

Thank you